

ETRG-0: Request for Refutation

The adversarial review packet — S/E/P/Q claims, checkpoints, and kill criteria (v0.2)

July 2026 · The ETRG Program

Provenance disclosure: this document was constructed by four AI models (Claude Fable 5 — synthesis and implementation; DeepSeek-V4-Pro — derivations; GLM-5.2 — refereeing; Kimi-K2.7 — numerics) in an adversarial-review protocol, with a human originator (B. Kloosterman) supplying the thesis and direction. Every claim's attack history, including retractions by all four models, is preserved in the accompanying repository. Numerical results are reproducible from committed scripts. Treat accordingly.

ETRG-0: Request for Refutation

An adversarial review packet for a hypothesis stitching entropic time to entanglement gravity. *Version 0.2 — July 2026. Self-contained; no prior context required. v0.1 → v0.2: after round 1 (three blind constructive passes + three adversarial reviews, all AI, July 2026): S3's lock is now derived in weakened, precise form in the companion `ETRG-0_lock_note.md` (v0.2, with adjudication appendix); checkpoint Q10 added; Q7 upgraded from open worry to quantified deviation scale; references extended (Padmanabhan, unimodular gravity, Hawking–Ellis).*

How to use this document

You are being asked to attack a hypothesis, not to admire it. Every claim below is numbered. The most useful response is “claim Sn fails because X” or “checkpoint Qn is already answered in the negative by [paper].” The second most useful response is “S-claims 1–5 were already proposed by [author, year] and refuted by [author, year]” — prior-art discovery is a win, not an embarrassment. Equally welcome is the constructive challenge near the end: derive your *own* competing entropic construction from the cited experiment and report where it lands. Vague encouragement is the least useful response.

Provenance disclosure: this framework was assembled in an exploratory research session between a human (background: ML/computer-vision engineering, not professional physics) and an AI assistant (Claude, Anthropic), building on a 2026 cold-atom experiment and the entanglement-geometry literature. All E-claims below belong to the cited literature. The S-claims and P-claims are the session's synthesis and carry no authority beyond their arguments. Numerical checks were run in companion code (exact diagonalization and free-fermion lattice computations; available on request). Treat accordingly.

The hypothesis in three sentences

There is one fundamental currency — entropy relative to a partition of a globally stationary quantum state — and both time and gravity are exchange rates of that currency. Coarse-grained entropy exchange between subsystems supplies the flow, rate, and arrow of time (as recently realized experimentally in a cold-atom “mini-universe”); fine-grained entanglement entropy across surfaces supplies spatial geometry via the entanglement first law (the Jacobson/Faulkner mechanism). The two faces provably coincide at causal horizons (Bisognano–Wichmann), which locks the temporal and spatial metric sectors together (PPN $\gamma = 1$) and makes the famous factor-of-two in relativistic light bending the direct observational readout of the lock.

E-claims: established results the hypothesis imports

Attack these only if we have misstated them.

1. **E1.** Canonical quantum gravity yields the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$; no external time parameter exists (DeWitt 1967; reviews: Isham 1992, Kuchař 1992; Henneaux, Collège de France lectures 2024–25).
2. **E2.** Relational/conditional-probability time constructions exist and, in modern covariant form, evade the classic propagator objection (Page–Wootters 1983; Giovannetti, Lloyd, Maccone PRD 2015).
3. **E3.** A laboratory realization now exists: a BEC partitioned into observed/unobserved sectors, with an entropic time $\tau = (\sigma/k_B) / (dS/d\phi) |d\phi|$ that orders repeated expansion/recollapse cycles, and an effective Schrödinger equation parameterized by τ that reproduces the data (Barontini, Phys. Rev. Research 2026, arXiv:2509.07745).
4. **E4.** Physical clocks require entropy production; tick resolution is thermodynamically bounded (Erker et al., PRX 2017, and successors).
5. **E5.** In stationary equilibrium in a gravitational field, $T(x)\sqrt{-g_{00}(x)} = \text{const}$ (Tolman 1930; Tolman–Ehrenfest 1930). Thermal/modular time is definable for generic states via Tomita–Takesaki theory (Connes–Rovelli 1994; Rovelli–Smerlak 2011).
6. **E6.** The vacuum restricted to a Rindler wedge is thermal; its modular flow is the boost (Bisognano–Wichmann 1976), giving the Unruh temperature $k_{BT} = \hbar a / 2\pi c$ (Unruh 1976).
7. **E7.** The first law of entanglement $\delta S_A = \delta\langle K_A \rangle$ holds exactly to first order; applied to small causal diamonds with the area-law calibration $S = A/4G\hbar$ it yields the linearized Einstein equations (Faulkner, Guica, Hartman, Myers, Van Raamsdonk 2014) and, with an “entanglement equilibrium” postulate, the full nonlinear ones (Jacobson 2015).
8. **E8.** In the standard PPN weak field, transverse acceleration is $a_\perp = -(1 + \gamma v^2/c^2) \nabla_\perp \Phi$; slow matter is γ -blind, light feels $(1+\gamma)$; measured deflection $1.75''$ at the solar limb requires $\gamma = 1$, and Cassini gives $|\gamma-1| \lesssim 2.3 \times 10^{-5}$ (Bertotti et al. 2003). A pure time-dilation (scalar) theory predicts half the bending (Einstein 1911; Nordström’s consistent completion bends light not at all).
9. **E9.** Gravity enters neutron and atom interferometry as a clean unitary phase (COW 1975; qBounce). This falsifies gravity mechanisms based on coarse-grained/dissipative entropy (Kobakhidze 2011 contra Verlinde 2011) but not unitary ones.
10. **E10.** GW170817 constrains gravitational-wave and electromagnetic propagation speeds to agree within $\sim 10^{-15}$ (Abbott et al. 2017).
11. **E11.** Lattice fact used in our checks: the vacuum correlation kernel of an interval (free fermions, half filling) commutes exactly with a tridiagonal operator whose hopping weights follow the parabola $(i+1)(\ell-1-i)/2$ — the discrete avatar of the Rindler boost weight $\beta(x) \propto (R^2-x^2)/2R$ (Slepian’s commuting-operator structure; entanglement-Hamiltonian context: Eisler–Peschel 2017).

S-claims: the actual hypothesis (attack here)

1. **S1 (One currency).** Coarse-grained entropic time (E3-type) and fine-grained entanglement geometry (E7-type) are two operational faces of a single modular structure, coinciding exactly at causal horizons via E6. Neither is more fundamental; the partition determines which face an observer reads.
2. **S2 (Lapse postulate).** The time–time metric component is *defined* by the local exchange rate between proper time and global entropic/thermal time: $\sqrt{-g_{00}(x)} = d\tau_{\text{proper}}/dt$, reducing in equilibrium to the inverse Tolman–Ehrenfest relation and generalizing to arbitrary states via modular flow (E5).

Newtonian gravity is then the maximal-aging response to the spatial variation of this clock-rate field: $a = -c^2 \nabla \ln \sqrt{-g_{00}}$.

3. **S3 (Geometry closure and the factor of two).** The spatial metric sector is fixed independently by the entanglement first law (E7), importing Einstein's equations and forcing $\gamma = 1$. Because both sectors descend from one boost/modular generator (E6, E11), their equality is structural, not tuned. The doubled deflection of light relative to slow matter — the $(1+\gamma v^2/c^2)$ law of E8 — is therefore the direct observational readout of S1's lock: a particle's bending ratio measures the fraction of its worldline spent sampling space versus time. *The precise, post-round-2 form of this claim is in `ETRG-0_lock_note.md`: the entropic input is algebraically the trace-free (unimodular) sector of the Einstein equations (published foundation: Alonso-Serrano & Liška 2022); light bending reads exactly that sector; the lock of the temporal face to the spatial face is theorem-grade at entanglement equilibrium, with dimensionless deviation $\delta_{\text{lock}} = S(\rho||\rho_{\text{vac}})/S_{\text{BH}} \leq 2G\Delta E/(c^4 R)$ beyond it, and `ETRG-0_Q10_note.md` supplies the partition-selection mechanism (dynamical consistency + Takesaki) at horizons.*
4. **S4 (Internal lapse).** Observed-sector dynamics obeys $i\hbar d|\psi\rangle/dt = \tilde{N}[\psi] \hat{H}|\psi\rangle$ with $\tilde{N}[\psi] = |dS[\psi]/dt|^{-1}$ computable from the instantaneous state alone. Stasis points ($dS \rightarrow 0$) are coordinate singularities of entropic time: the lapse diverges and duration information is internally irrecoverable there.
5. **S5 (Universality and unitarity).** Null cones are the causal structure of entanglement propagation, common to all massless excitations (consistent with E10); and because the gravitating entropy is fine-grained, gravity introduces no decoherence channel (consistent with E9).

P-claims: predictions

1. **P1.** In any Barontini-class system, total accumulated τ collapses smoothly to zero with the sector coupling (barrier-height sweep), the lapse is expressible from internal data alone, and reconstruction errors concentrate at stasis points. Testable on the existing Birmingham apparatus.
2. **P2 (the discriminating experiment).** In a flowing-BEC sonic-horizon setup (Steinhauer geometry), three independently measurable quantities must lock through one modular rate: the Hawking phonon temperature (density–density correlations), the entanglement entropy across the horizon (area law with 1D log correction, coefficient set by the analogue surface gravity), and the interior sector's entropic-time lapse. A measured violation of the lock falsifies S1. *Round-3 sharpening (supersedes the round-2 version, which lattice adjudication refuted): the lock discriminates by **which leg moves**, not by total violation magnitude. A coherent (spectrum-preserving) drive of the condensate leaves the entanglement-entropy leg exactly unchanged ($\Delta S_{\text{ent}} = 0$) while contributing to the modular-energy leg at linear order in energy; an incoherent excitation of identical energy at the same location moves the entropy leg at linear order while its relative entropy is suppressed to second order — the first law itself. Verified on a free-boson lattice (`coherent_thermal_check.py`: S_{rel} slopes 1.000 vs ≈ 2 ; ΔS_{ent} exactly 0 vs linear). Falsifiable signature: the leg-resolved response to coherent vs incoherent injections of matched energy must follow this pattern; any other ordering falsifies the modular mechanism specifically.*

3. **P3.** No gravitationally induced decoherence at any interferometric precision — a standing bet against Diósi–Penrose-type collapse and against dissipative entropic gravity. A confirmed detection of gravity-correlated dephasing falsifies S5.

Q-checkpoints: where to kill it without a laboratory

These are consistency requirements the hypothesis has **not** yet passed. A negative proof at any of them is fatal or near-fatal; that is the point of listing them.

1. **Q1 (Constraint algebra).** Any candidate gravity theory must reproduce the Dirac hypersurface-deformation algebra (or explain its effective emergence). Does the S2+S3 construction close the algebra? A proof that it cannot ends the program.
2. **Q2 (Out-of-equilibrium lapse).** S2's equilibrium anchor is Tolman–Ehrenfest; the generic-state extension leans on modular flow, which for arbitrary states and regions is non-local and non-geometric. Is the lapse well-defined, covariant, and approximately geometric for realistic non-equilibrium states — e.g., a collapsing star? If not, S2 is vacuous outside static spacetimes.
3. **Q3 (Factorization covariance).** Nothing in $\hat{H}|\Psi\rangle = 0$ selects the bright/dark split. Do the physical predictions depend on the choice of tensor factorization, and can entropic time be gamed by re-partitioning (the “clock ambiguity,” Albrecht–Iglesias 2008)? If τ is factorization-dependent in observable ways, S1 and S4 are empty.
4. **Q4 (Kuchař's objections).** The multiple-choice, functional-evolution, and Hilbert-space problems of internal time (Kuchař 1992). GLM 2015 addresses the propagator objection for Page–Wootters; does the *entropic* clock variant inherit the fix, or does using a thermodynamic (non-canonical) clock variable reopen the wounds?
5. **Q5 (Nonlinear-lapse pathologies).** $\tilde{N}[\psi]$ makes the τ -evolution nonlinear in the state. We argue this is a pure reparameterization — the state-space trajectory is identical to linear QM, only the time label is state-dependent — and hence evades Gisin-type superluminal-signaling theorems. Prove or refute: can two agents sharing entanglement exploit state-dependent time labels to signal?
6. **Q6 (Exact Lorentz invariance).** If null cones are emergent (Lieb–Robinson-like), existing bounds bite hard: Fermi GRB time-of-flight excludes linear Planck-scale dispersion; UHECR constraints go further. Is there a continuum limit with *exact* LI, or is the framework already excluded by dispersion bounds? Note this objection applies to the substrate reading of S5, not to the phenomenological layer, which imports exact cones.
7. **Q7 (Locking away from horizons).** E6 makes modular flow geometric only for wedges and conformal diamonds. Away from these, the two sectors could in principle decouple, predicting PPN deviations where none are observed. Quantify: does the S1 lock degrade for generic regions, and if so, at what observable level?
8. **Q8 (de Sitter and Λ).** The E7 machinery is AdS-native; our universe has $\Lambda > 0$. Is a positive- Λ extension possible even in principle within this construction, or does S3 silently assume a boundary structure our universe lacks?

9. **Q9 (Novel content).** Outside P1–P3, is there any regime where the hypothesis differs from GR+QM? If the honest answer is no, state plainly whether this is a physical theory, a reinterpretation, or a research heuristic — and whether that verdict is fatal. *Round-1 partial answer: in the weak field the framework is exactly linearized GR's trace-free sector — no new predictions there, by construction; the candidate novel regimes are L5's off-equilibrium deviation scale ($\Delta E \cdot \ell^2_{\text{Planck}}/A$) and the P1–P3 laboratory seams.*
10. **Q10 (Two partitions, one generator — found in round 1).** The temporal sector's operational partition (Barontini's coarse-grained observed/unobserved split) and the spatial sector's partition (region/complement across a surface) are different tensor factorizations of the Hilbert space, with different modular generators in general. S1/S3 and the lock note's L5 require them to define the *same* modular structure at causal horizons. Bisognano–Wichmann makes this plausible (the wedge algebra underwrites both) but no proof exists. If the two factorizations fail to coincide even at horizons, the two faces of the lock belong to two different operators and the structural argument for $\gamma = 1$ collapses — this is now the sharpest known internal vulnerability.

Anticipated objections, steelmanned

“Entropic gravity was refuted by neutron interferometry.” Correct for coarse-grained, dissipative mechanisms (E9). Here the gravitating entropy is fine-grained and the dynamics unitary; the refutation is imported as a design constraint, and P3 doubles down on it. Attack S5's unitarity claim instead.

“The factor of two is just standard GR.” Also correct — no new number is claimed. The claim (S3) is structural: *why* the temporal and spatial sectors are equal, namely one shared modular generator. Refute the explanation: show the two sectors need not be locked (Q7), or find prior art that makes the identical argument.

“Page–Wootters conditional probabilities give wrong propagators.” The 2015 covariant reformulation resolves this, and E3 is an operational realization. The open flank is Q4: whether an *entropic* clock variable, which is not a canonical conjugate of anything, inherits the resolution.

“Nonlinear quantum mechanics implies signaling.” The nonlinearity lives in the time parameter, not the dynamics; see Q5 for the demanded proof.

“Unfalsifiable dressing on GR+QM.” P2 is a quantitative three-way lock that can fail; P3 is a standing decoherence bet; Q1, Q3, and Q6 are armchair falsifiers. But Q9 concedes the general worry is the most serious one on the list.

Kill criteria

We commit in advance to treating the following as fatal: a proof that the constraint algebra cannot close (Q1); a demonstration that predictions depend on the factorization choice (Q3); a working signaling protocol from the state-dependent lapse (Q5); a proof that the coarse-grained and region/complement factorizations cannot define the same modular structure at causal horizons (Q10); measured violation of the P2 lock; confirmed

gravitational decoherence (P3); or discovery of prior art proposing S1–S4 together with a standing refutation. We commit to treating Q9 answered “no novel content, ever, in principle” as a demotion from hypothesis to heuristic.

Numerical evidence to date (weak but nonzero)

Exact-diagonalization checks ($N = 120$ two-mode Bose–Hubbard analogue of E3): entropic time orders bang/crunch cycles, accumulating $\tau = 90.5$ with sector coupling on versus 0.61 with the barrier raised; the S4 internal-lapse Schrödinger equation, using no external clock, reproduces the dynamics with RMS error 0.058 and recovers 98.3% of total duration, with errors concentrated at stasis points as predicted. A 2D matter wave in a pure clock-rate field falls on the Newtonian parabola to 2×10^{-4} (S2 mechanism). On a free-fermion chain, the first law $\delta S_A = \delta \langle K_A \rangle$ verifies with signal slope 1.00 and remainder slope 2.00 over three decades, and the parabolic boost commutes with the vacuum interval to $\|[\rho_A, T]\| \approx 10^{-17}$ against 10^{-2} for a uniform-weight control (E7/E11 pillars). These validate the imported machinery on lattices; they do not validate the stitching claims S1–S3, which is precisely why this document exists.

The constructive challenge (explicitly requested)

Beyond attacking the claims above, please attempt the following independent derivation — ideally *before* reading S1–S5, if your workflow allows a blind pass.

Starting only from Barontini’s experimental construction (E3: a closed quantum system partitioned into observed and unobserved sectors, with time defined by entropy exchange; arXiv:2509.07745), **build your own entropic theory of time and gravity**. Your construction must, at minimum: recover Newtonian gravity for slow matter; explain why particles moving at c curve into a gravity well at exactly twice the rate of particles at rest — the measured $1.75''$ solar deflection versus the $0.87''$ of any pure time-dilation theory, i.e. PPN $\gamma = 1$ to Cassini precision (E8); preserve the unitary gravitational phase seen in interferometry (E9); and put all massless species on one null cone (E10).

The factor of two is the designed choke point of this exercise. Any mechanism that produces gravity purely as differential time flow is a scalar theory and lands on $0.87''$ — Nordström’s grave. A successful construction must therefore generate a *spatial* metric sector from entropic or informational ingredients as well, and the diagnostic question is which mechanism you reach for and how the two sectors get locked at equal strength. Three outcomes are each valuable: (i) independent convergence on the ETRG-0 stitching — entanglement first law supplying g_{ij} , a shared modular/boost structure enforcing the lock — which would be evidence the assembly is natural rather than contrived; (ii) a different mechanism that passes the same gates, i.e. a genuine competitor we want to hear about; (iii) an argument that *no* entropic construction can produce $\gamma = 1$ without importing Einstein’s equations wholesale — which would demote the entire program, ETRG-0 included, and deserves a theorem sketch if you can manage one. Whichever lands, please conclude by comparing your construction against S1–S5 and stating explicitly where they agree, where they differ, and which you consider better motivated.

Specific requests to AI reviewers

Complete the constructive challenge above first if you can, and report convergence or divergence with S1–S5. Then verify each E-claim as stated against your knowledge and, where able, the primary sources. Search for prior art: any framework combining thermal/entropic time (Connes–Rovelli lineage) with entanglement-first-law gravity (Jacobson lineage) as a single construction, and any published refutation of such a combination. Attempt Q1–Q9 in order of expected lethality (our guess: Q3, Q2, Q1). Check the E8 equation of motion and the S3 interpretation of the deflection factor for errors. Propose the cheapest experiment or calculation that would separate this hypothesis from pure GR+QM faster than P2.

Core references

DeWitt, Phys. Rev. 160, 1113 (1967) · Page & Wootters, PRD 27, 2885 (1983) · Kuchař, “Time and interpretations of quantum gravity” (1992) · Isham, “Canonical quantum gravity and the problem of time” (1992) · Connes & Rovelli, Class. Quantum Grav. 11, 2899 (1994) · Giovannetti, Lloyd, Maccone, PRD 92, 045033 (2015) · Barontini, Phys. Rev. Research (2026), arXiv:2509.07745 · Erker et al., PRX 7, 031022 (2017) · Tolman & Ehrenfest, Phys. Rev. 36, 1791 (1930) · Bisognano & Wichmann, J. Math. Phys. 17, 303 (1976) · Unruh, PRD 14, 870 (1976) · Jacobson, PRL 75, 1260 (1995) · Jacobson, PRL 116, 201101 (2016) (“Entanglement equilibrium,” 2015 preprint) · Faulkner, Guica, Hartman, Myers, Van Raamsdonk, JHEP 03 (2014) 051 · Ryu & Takayanagi, PRL 96, 181602 (2006) · Van Raamsdonk, Gen. Rel. Grav. 42, 2323 (2010) · Maldacena & Susskind, Fortschr. Phys. 61, 781 (2013) · Verlinde, JHEP 04 (2011) 029; SciPost Phys. 2, 016 (2017) · Kobakhidze, PRD 83, 021502 (2011) · Colella, Overhauser, Werner, PRL 34, 1472 (1975) · Abbott et al., ApJL 848, L13 (2017) · Steinhauer, Nat. Phys. 12, 959 (2016) · Eisler & Peschel, J. Phys. A 50, 284003 (2017) · Bertotti, Iess, Tortora, Nature 425, 374 (2003) · Albrecht & Iglesias, PRD 77, 063506 (2008) · Padmanabhan, Rep. Prog. Phys. 73, 046901 (2010) · Padmanabhan, Gen. Rel. Grav. 46, 1673 (2014) · Hawking & Ellis, *The Large Scale Structure of Space-Time* (1973), §4.3 · Weinberg, Rev. Mod. Phys. 61, 1 (1989) · Ellis, van Elst, Murugan, Uzan, Class. Quantum Grav. 28, 225007 (2011) · Eddington, *Space, Time and Gravitation* (1920).

Companion materials: ETRG-0.md (full axiomatization), etrg_demos.py (all numerical checks), three figures. Ask the circulator for copies.